

MOHAMED JRIDI

Düsseldorf, Germany

LinkedIn: [in/medjridi/](https://www.linkedin.com/in/medjridi/) | Github: github.com/moodjx



About me: Engineer focused on solving complex problems of scaling event-driven systems. My background is rooted in Java/Python web and IoT architectures.

Core Focus: Python/Java Ecosystems • IoT Development • Agile Frameworks • AI Intergration

Professional Experience

Freelancer - Düsseldorf

09/2024 - Ongoing

- Built landing pages for small businesses.
- Built projects using AI Agents, including ChatBots and Browser Extensions (Plugins).
- AI Agents development lifecycle, including function calling with MCP, Tool chaining, guardrails and Context Engineering.
- Achieved significant progress in the German language.

Lead IoT Developer - Livello GmbH, Düsseldorf

11/2023 - 08/2024

Scaling a cloud-to-edge platform for more markets while building the engineering culture.

- Team Mentorship: Supported a team of 5 developers through code reviews and pair programming, fostering a culture of clean code and shared technical ownership.
- Re-architected a growing IoT software stack in Python, Java, Typescript, MQTT using a event-based services approach.
- Cross-functionality: Worked with the computer vision team on an ML counting system, in the back-end on architecture and DevOps, and code and the hardware team on device integrations.
- Legacy Fleet Overhaul: Refurbished the existing device fleet by porting modern IoT capabilities to legacy hardware, extending the asset lifecycle.
- Device Integrations: Worked on OTA for the devices and added RFID labels, an NFC device and new payment devices to the main product.
- Customer Support: Technical assistance and support.

Software Engineer - IoT - Livello GmbH, Düsseldorf

04/2022 - 10/2023

Goal: New features and heavy maintenance. Strengthening system architecture and automating the management of a large-scale device fleet.

- Fleet Automation: Release and update automation with CI/CD pipelines in GitLab and Ansible.
- Hardware: Integrated new payment devices that work with the ZVT protocol and engineered a failover connectivity system using LTE modules to ensure constant device uptime.
- Release and update automation with CI/CD pipelines in GitLab and Ansible. Deployed a Prometheus/Grafana stack on edge devices, allowing the team to identify and resolve issues beforehand.

Internship - Livello GmbH, Düsseldorf

09/2021 - 03/2022

Software Developer in IoT, maintaining software, resolving bugs and a focus on solving a payment process speed and purchase flow.

Bachelor Thesis, Capstone Project - WorldSoft Group, Tunis

02/2018 - 06/2018

Full-stack Developer, I developed a Java/Angular web application for B2B car rental business.

Education

Master's Degree - Engineering studies, branch of Internet of Things Systems and Services (IoSyS), ESPRIT School of Engineering

Exchange Student at Fachhochschule Schmalkalden, Thuringen - 2020

- Program: 6 Months
- Software Engineering/Development program
- All Scored from 1 to 1,7

Bachelor's Degree in Fundamental Computer Science, FSB Tunisia - 2018

Abitur (Baccalauréat) in Computer Science, Sehr gut (Mention Très Bien)

Main Technical Skills

Programming & Languages

Python, Java, C/C++ (Arduino IDE), Javascript/TypeScript

Frameworks/Libraries: Spring Boot, Vert.x, Node.js, FastAPI, FastMCP, Google genai, aiogram

Cloud & IoT Platforms

Google Cloud Platform (GCP), Docker, Kubernetes, OAuth2, ClearBlade

DevOps & Tooling

CI/CD, Ansible, Prometheus, Grafana, Github, GitLab

Message Brokers

AMQP (RabbitMQ), MQTT (Mosquitto)

Databases

MongoDB, Redis, PostgreSQL

Systems & Networking

Linux System Administration, ARM Architecture, OpenVPN, Arch Linux

IoT Protocols

MQTT, AMQP, BLE, ZWT, MDB, LTE, NFC, RFID

Development Boards

Raspberry Pi 3/4B/4B+/Pico, Arduino, STM32

Methods & Architecture

Agile/Scrum, Microservices, Event-Driven Architecture, Reactive Programming, BDD, TDD, UML

Additional Skills

Artificial Intelligence

LLM APIs (Gemini, DeepSeek, Mistral), MCP, PydanticAI, Context Engineering, RAG

Tools & Development

Jira, Confluence, VS Code, Next-Gen AI Code Tools

Languages

Arabic: Mother tongue

German: B1

English: Fluent

French: Fluent

Hobbies

Reading

Cooking

Football

Calisthenics